University of Liberal Arts Bangladesh

Department of Computer Science & Engineering

CSE 4417 IoT Theory

Lecture 01:

Introduction to the Course

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Class Code
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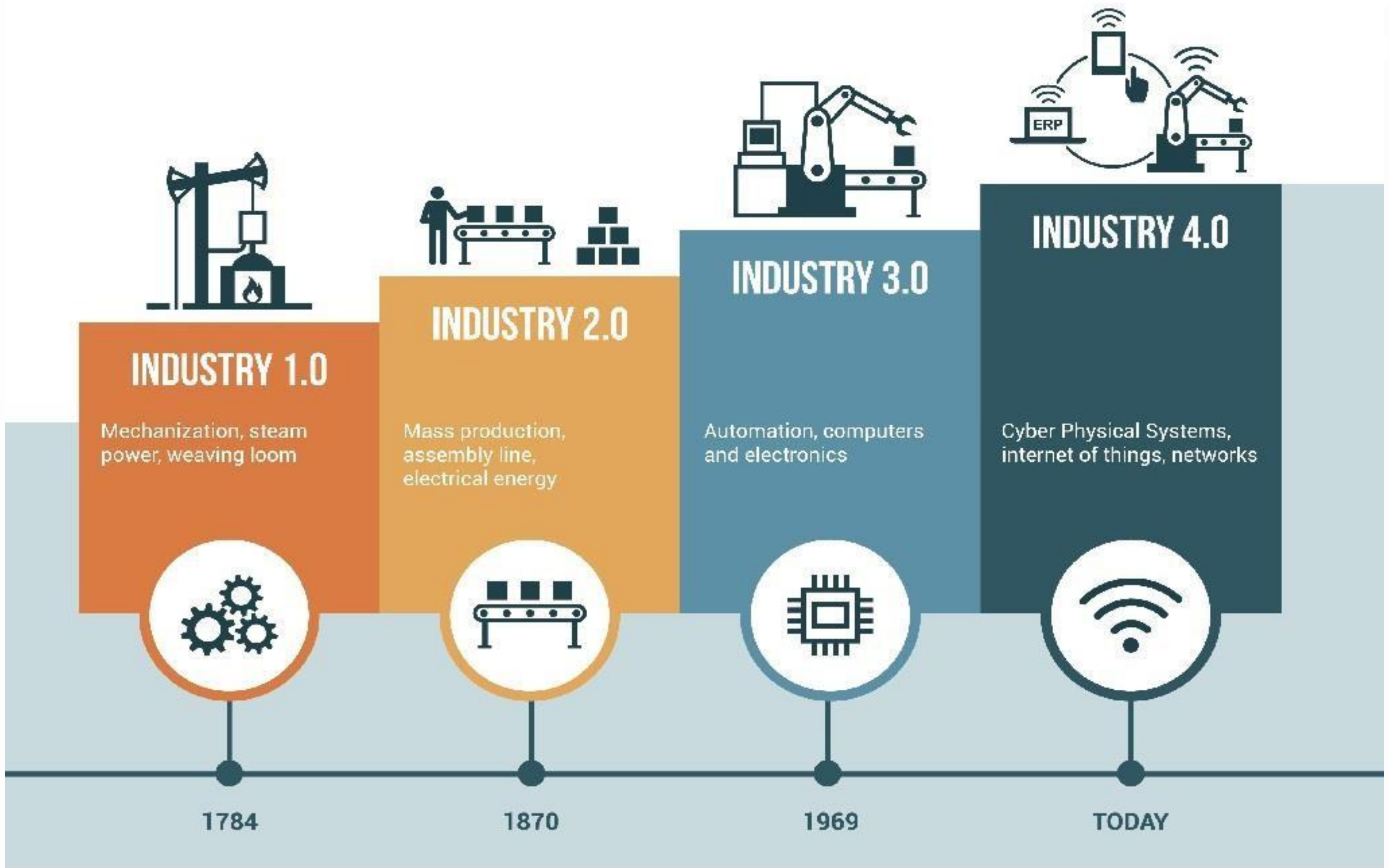
Objectives

- To discuss the fundamentals of Internet of Things (IoT)
- To discuss the functional architecture
- To introduce to the hardware component – Sensors and Actuators
- To discuss the various IoT based use cases

Fourth Industrial Revolution – 4IR



Source: Aberdeen Strategy Research [3]

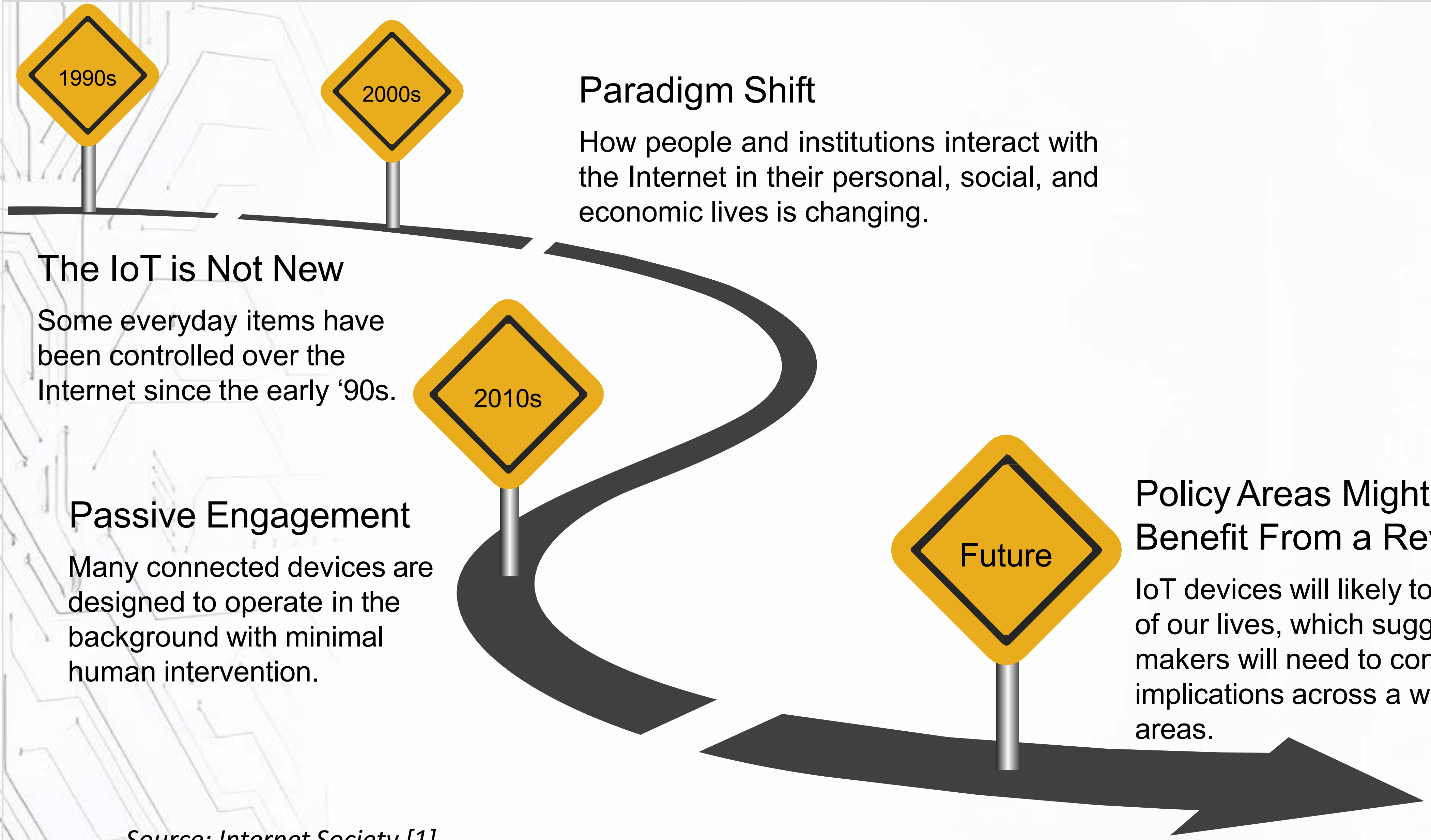
Internet of Things (IoT)

IoT is simply a concept wherein machines and everyday objects are connected via Internet

The *Thing* refers to all the things that can be connected to Internet

- Door locks
- Lights
- Household appliances
- Car Clothes

IoT – on the way



The IoT is Not New

Some everyday items have been controlled over the Internet since the early ‘90s.

Paradigm Shift

How people and institutions interact with the Internet in their personal, social, and economic lives is changing.

Passive Engagement

Many connected devices are designed to operate in the background with minimal human intervention.

Policy Areas Might Benefit From a Review

IoT devices will likely touch most aspects of our lives, which suggests policy makers will need to consider the broad implications across a wide array of areas.

Source: Internet Society [1]

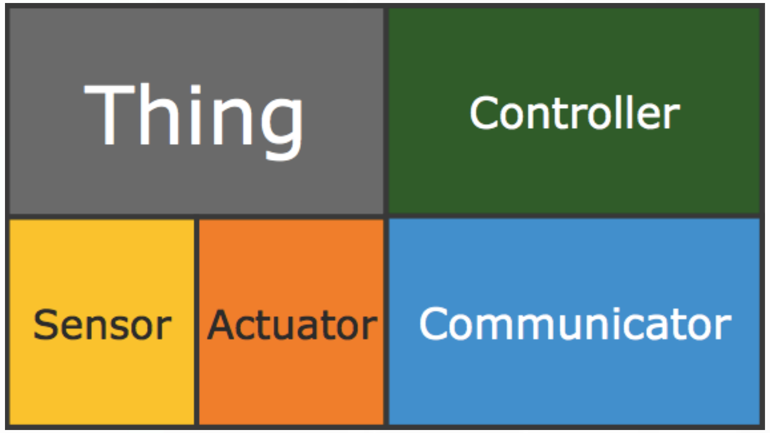
Connected Devices in the Past

- The concept of combining computers and networks to monitor and control devices has been around for decades
  - Systems for remotely monitoring meters on the electrical grid via telephone lines were in commercial use in late 1970
  - The first Internet device — an IP–enabled toaster that could be turned on and off over the Internet, was featured at an Internet conference in 1990

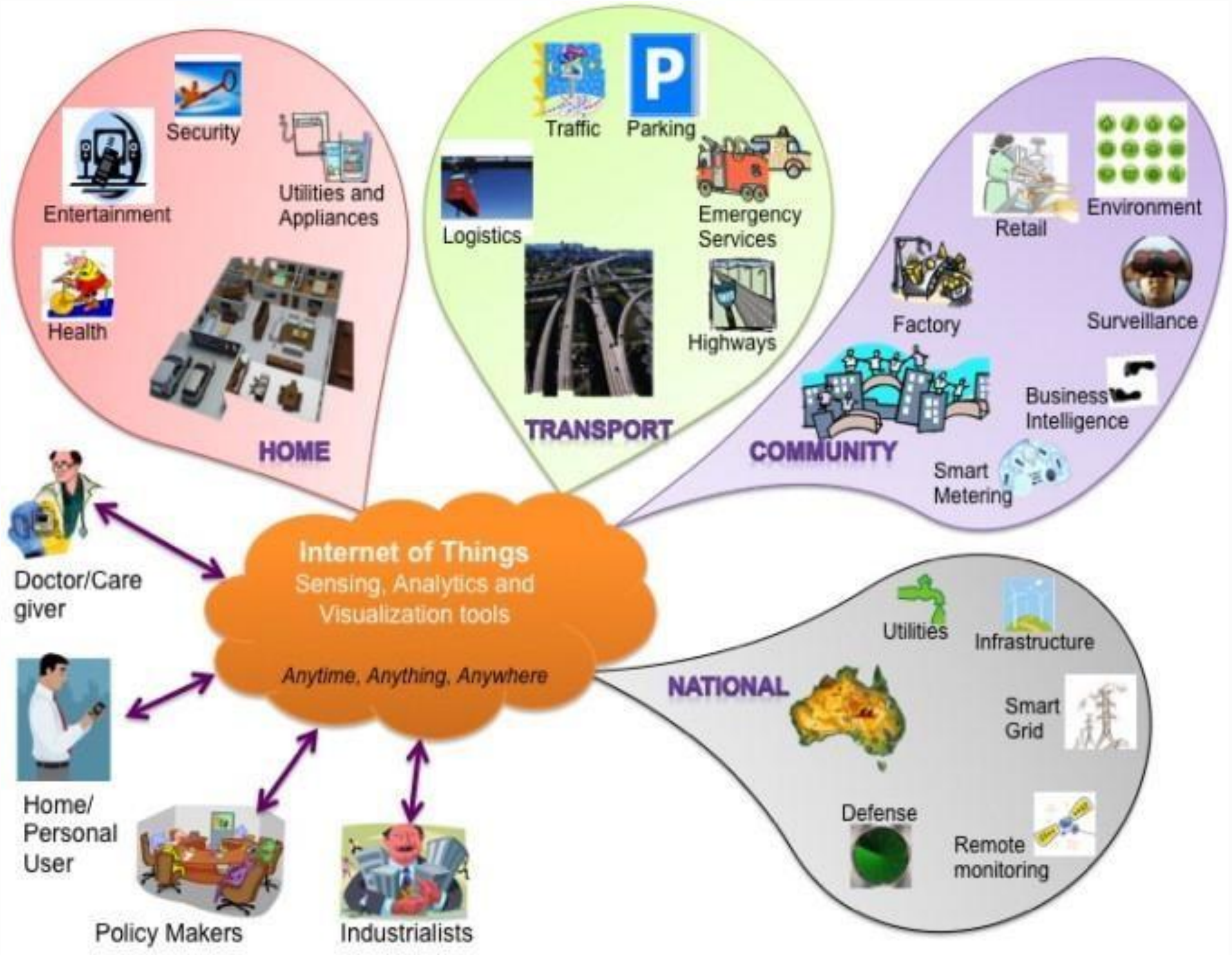
IoT

Things in IoT usually refers to IoT devices that:

- Exchange data with other connected devices and applications
- Send the data to centralized servers or cloud-based application for processing
- Perform some tasks locally and other tasks within the IoT infrastructure



IoT - Applications



IoT - Applications

McKinsey Global Institute describes the wide range of potential applications in terms of *settings* where IoT can benefits industry and users.

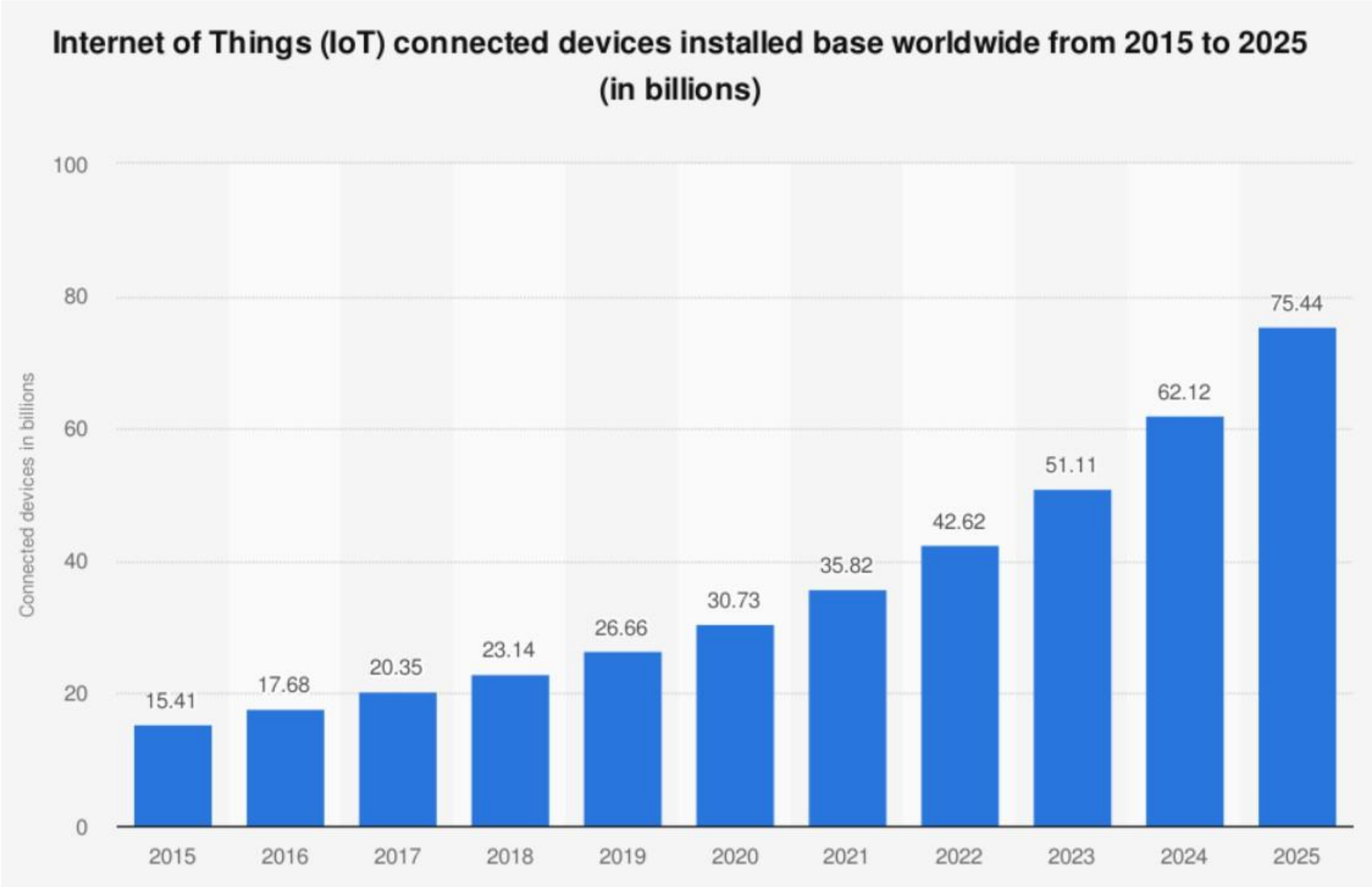
Setting	Description	Examples
Human	Devices attached or inside the human body	Devices to monitor and maintain human health and wellness; disease management, increased fitness, higher productivity
Home	Buildings where people live	Home controllers and security systems
Retail Environments	Spaces where consumers engage in commerce	Stores, banks, restaurants, arenas – anywhere consumers consider and buy; self-checkout, in-store offers, inventory optimization
Offices	Spaces where knowledge workers work	Energy management and security in office buildings; improved productivity, including for mobile employees
Factories	Standardized production environments	Places with repetitive work routines, including hospitals and farms; operating efficiencies, optimizing equipment use and inventory

IoT - Applications

McKinsey Global Institute describes the wide range of potential applications in terms of *settings* where IoT can benefits industry and users.

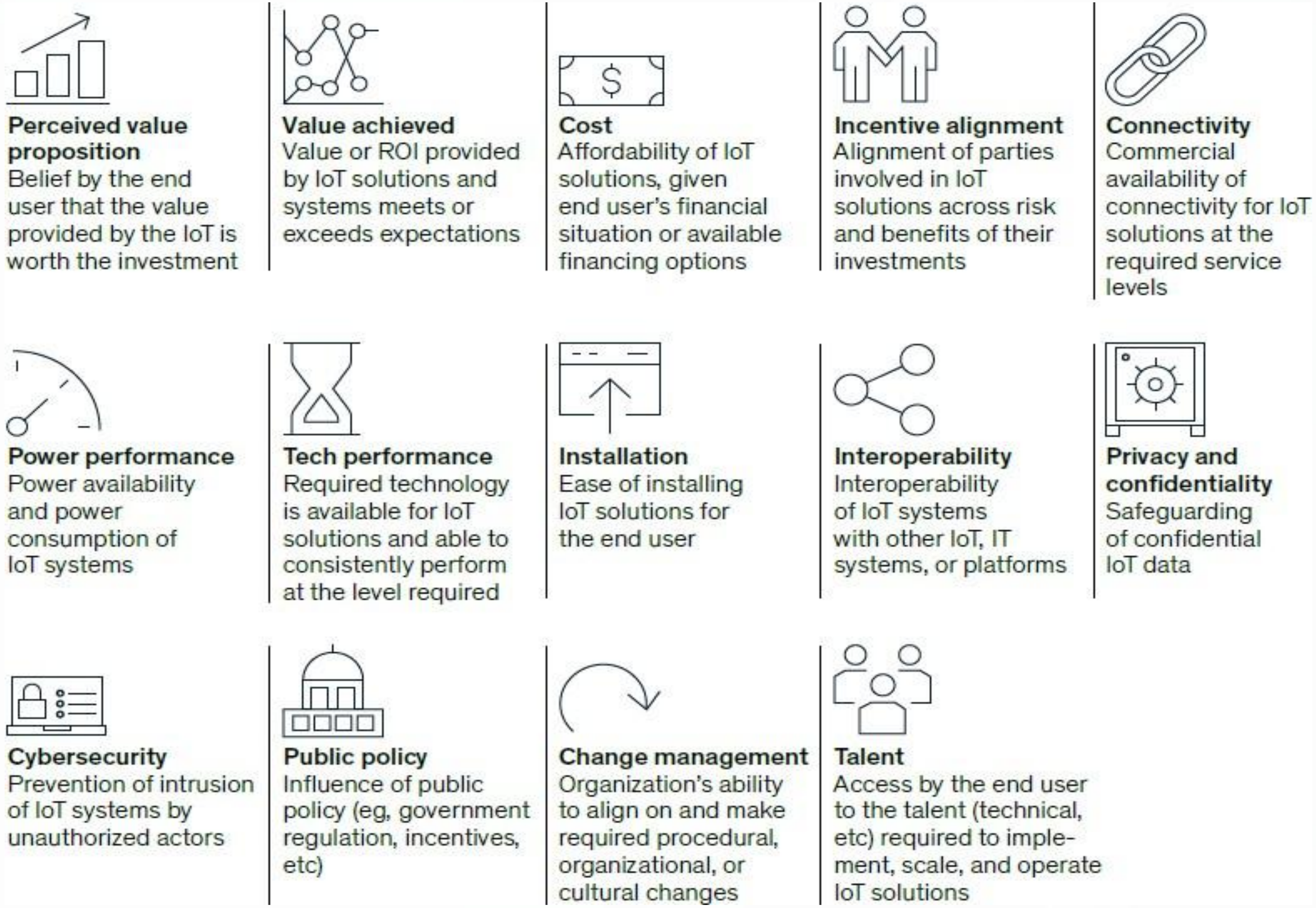
Setting	Description	Examples
Worksites	Custom production environments	Mining, oil and gas, construction; operating efficiencies, predictive maintenance, health and safety
Vehicles	Systems inside moving vehicles	Vehicles including cars, trucks, ships, aircraft, and trains; condition-based maintenance, usage-based design, pre-sales analytics
Cities	Urban environments	Public spaces and infrastructure in urban settings; adaptive traffic control, smart meters, environmental monitoring, resource management
Outside	Between urban environments (and outside other settings)	Outside uses include railroad tracks, autonomous vehicles (outside urban locations), and flight navigation; real-time routing, connected navigation, shipment tracking

Rise of connected devices



Source: Statista 2021 [2]

Factors Driving Adoption within IoT



Challenges

Security & Privacy

Ensuring security in IoT products and services. Maintaining transparency and fairness in user data collection and handling

Interoperability and Standards

Maintaining interoperability among the systems, using open and widely accepted standard and vocabulary

Transformation to new model

How to transform from existing process and models? How to integrate IoT into existing business that may require building from new.

Regulatory, Policy and Legal Issues

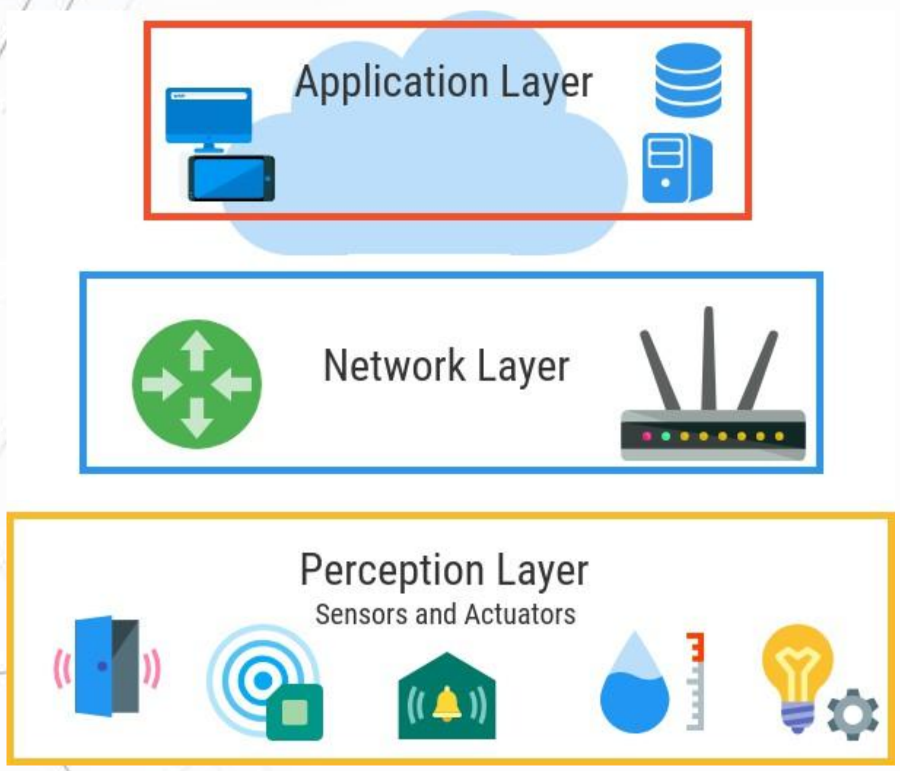
How to maintain the regularity, rights and policy issues to deal with the rapid rate of changes in IoT technology.

Emerging Economy and Sustainable Development

Delivering social and economic benefits to emerging economies, addressing the process of implementation, deployment, and the use of technology in less-developed regions.

Architecture of IoT

Layer-Wise Architecture



- The architecture of IoT is divided into three basic layers
  - Perception layer is used to collect data
  - Network layer provide data transmission services
  - Application layer deliver application specific services to users

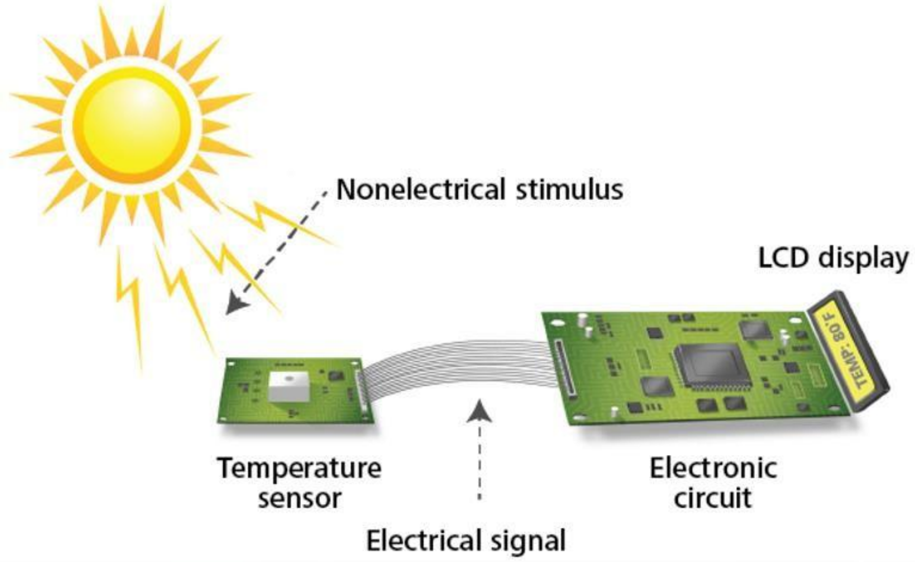
Functional View

CREATE	COMMUNICATE	AGGREGATE	ANALYZE	ACT
Sensors	*Network*	*Integration*	*Augmented intelligence*	*Augmented behavior*
Physical devices • Sensor devices	**Connectivity** • Processing elements • Connectivity elements • Connectivity frameworks	**Management** • IoT middleware **Accumulation** • Data ingestion • Data storage • Data abstraction **Existing data** • Unstructured data • Structured data	**Processing** • Processing engines/ frameworks • Data messaging **Analytics** • Stream analytics (data in motion) • Batch analytics, machine learning (data at rest)	**Applications** • User applications • Reporting **Edge computing** • Intelligent gateways • Fog computing platforms

Sensors

Sensors

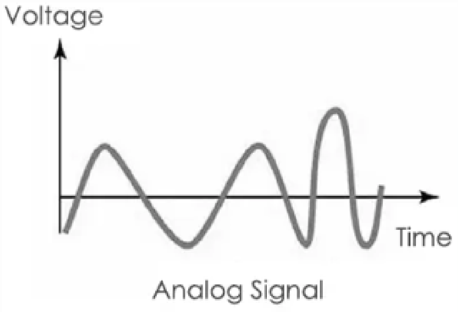
- Sensors perform some input functions by sensing or feeling the physical changes in characteristics of a system in response to a stimuli
- Converts a non-electrical input into an electrical signal that can be sent to an electronic circuit



Sensors

Analog sensors

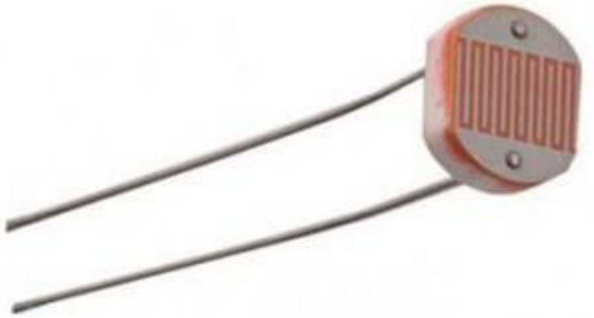
- Analogue sensors produce a continuous output signal or voltage which is generally proportional to the quantity being measured
- Analogue sensor senses the external parameters (wind speed, solar radiation, light intensity, etc.) and gives analog voltage as an output. Thus, The output voltage may be in the range of 0 to 5V. Logic High is "1" (3.5 to 5V) and Logic Low is "0" (0 to 3.5 V)



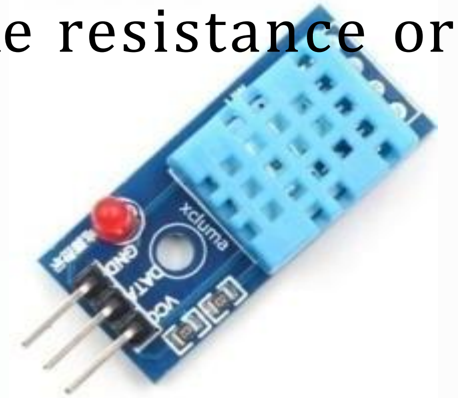
Sensors

Analog sensors (Example)

- Switch on and off loads automatically based on the day light incident on the Light dependent resistor (LDR)

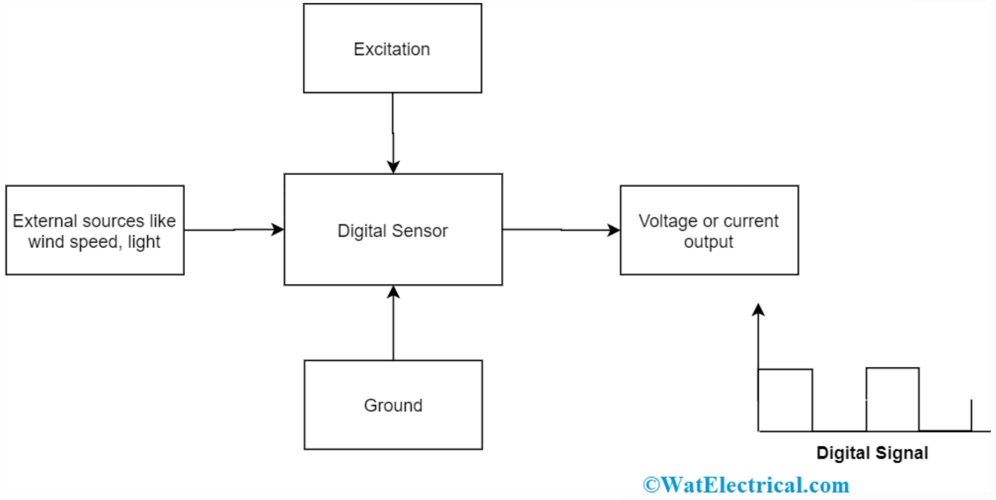


- Temperature Sensor, the changes in the Temperature correspond to change in its physical property like resistance or voltage.



Sensors

- Digital sensors
- Electrochemical or electrical sensors where the information is converted to digital form and then transmitted
- Digital sensors produce discrete values (0 and 1’s). Discrete values often called digital or binary signals in digital communication.

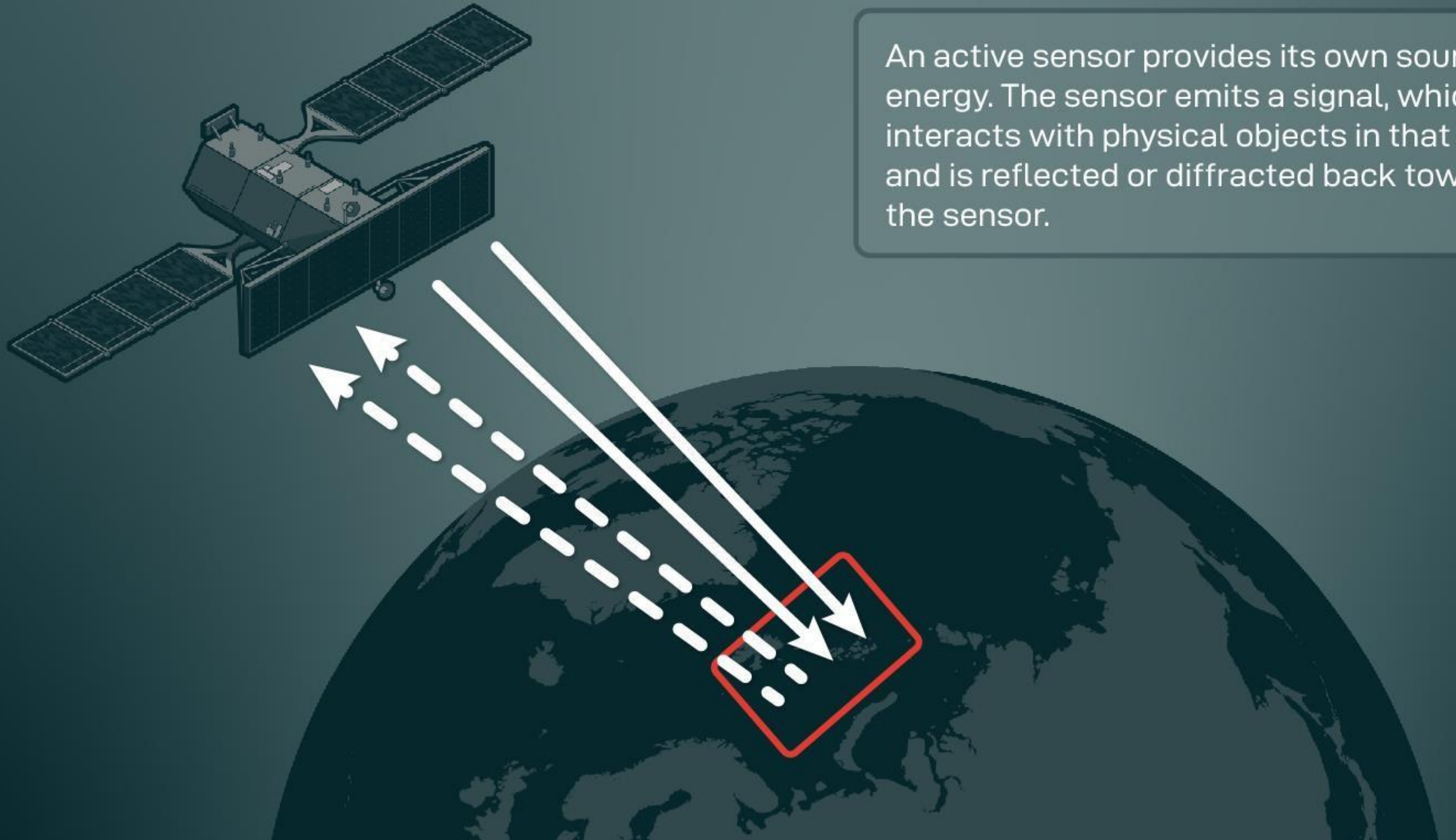


Sensors

- Digital sensors (Example)
- Digital Light Sensors can calculate intensity of light and provide a digital equivalent value.

- Temperature Sensor, output is a discrete digital value (usually,
- some numerical data after converting analog value to digital value)

Active Sensors

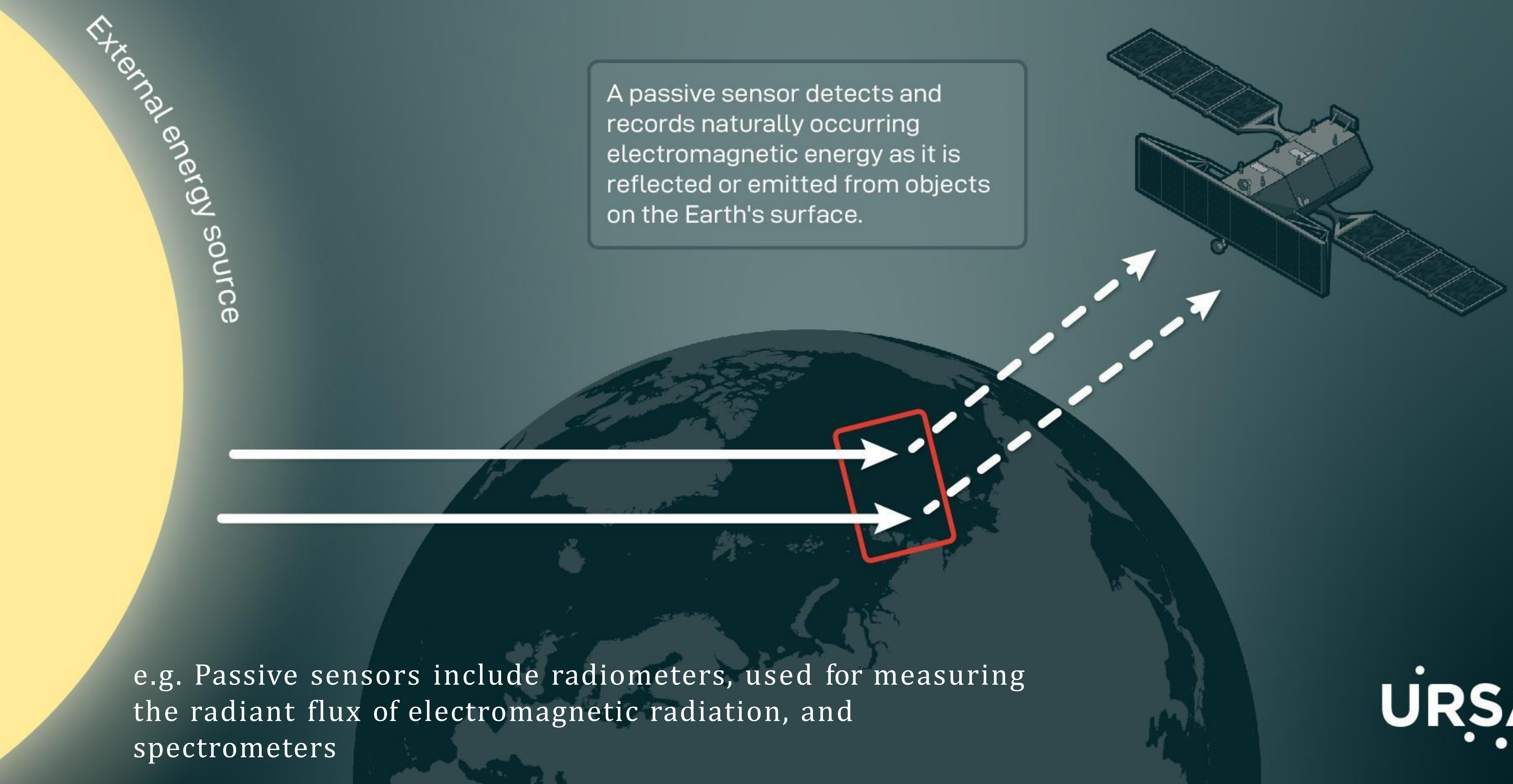


An active sensor provides its own source of energy. The sensor emits a signal, which interacts with physical objects in that area, and is reflected or diffracted back toward the sensor.

e.g. Active sensor-based technologies include: scanning electron microscopes, radar, GPS, x-ray

Passive Sensors

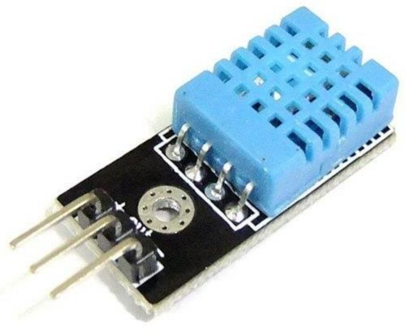
A passive sensor detects and records naturally occurring electromagnetic energy as it is reflected or emitted from objects on the Earth's surface.



e.g. Passive sensors include radiometers, used for measuring the radiant flux of electromagnetic radiation, and spectrometers

Sensors

Sensor types	Sensor description	Examples
Temperature	Measure the amount of heat or cold that is present in a system. Contact temperature sensors need to be in physical contact. Non-contact sensors do not need physical contact, they measure temperature through convection and radiation.	Thermometer Temperature
Chemical	Measure the concentration of chemicals in a system.	Smoke detector

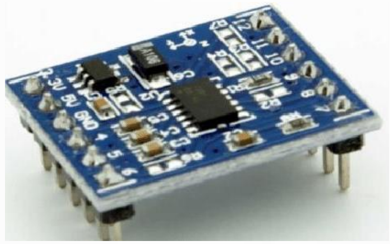


Sensors

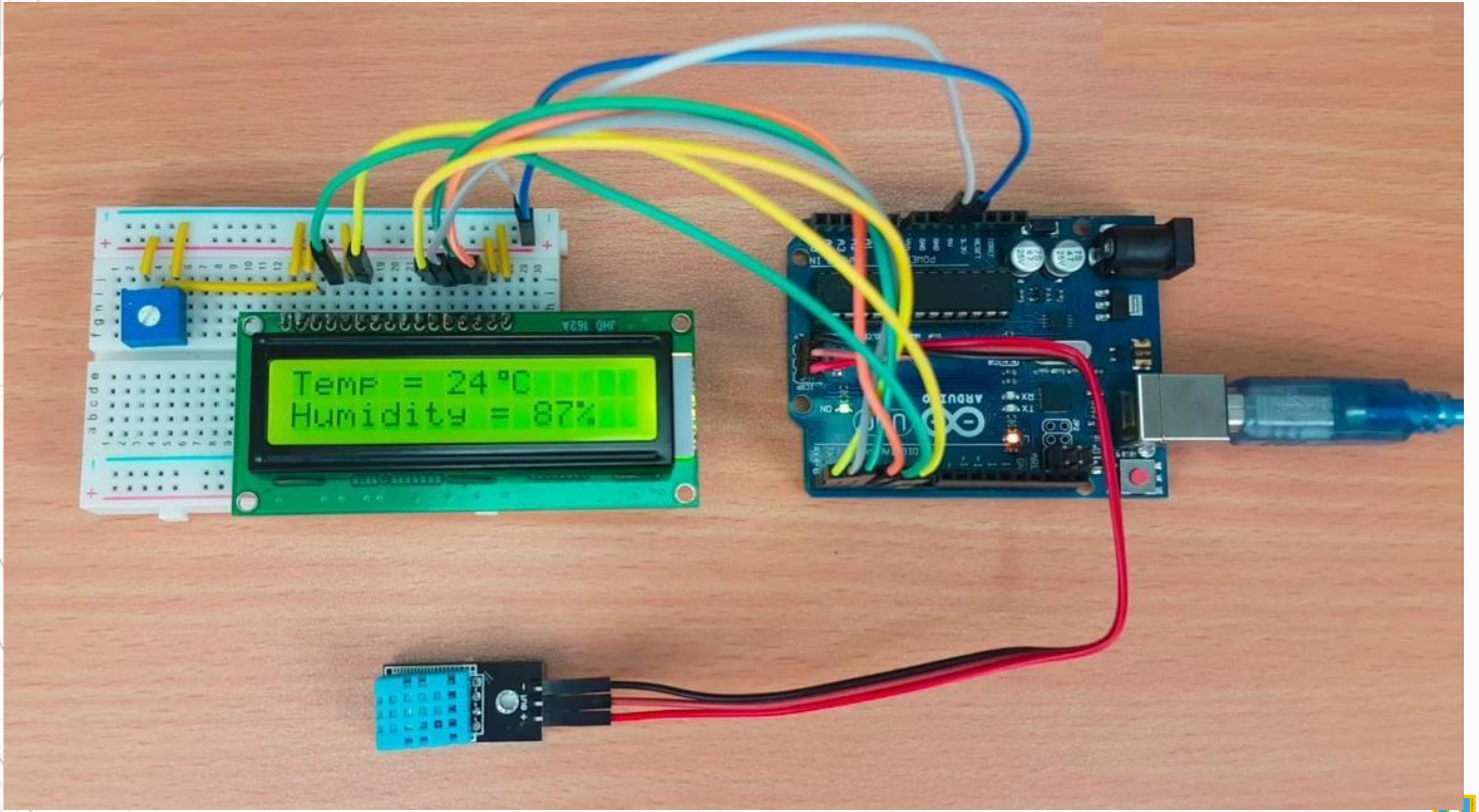
Sensor types	Sensor description	Examples
Flow	Flow sensors detect the rate of fluid flow. They measure the volume or velocity of fluid that has passed through a system in a given period of time.	Mass flow sensor Water meter
Humidity	Humidity sensors detect humidity (amount of water vapor) in the air or a mass.	Humidity Soil moisture sensor
Light	Light sensors detect the presence of light	Infrared sensor Photodetector

Sensors

Sensor types	Sensor description	Examples
Position	Measures the position of an object	Proximity sensor
Velocity and acceleration	Velocity sensors indicates how fast an object moves along a straight line or how fast it rotates. Acceleration sensors measure changes in velocity.	Gyroscope Accelerometer
Pressure	Related to force sensors and measure the force applied by liquids or gases.	Barometer

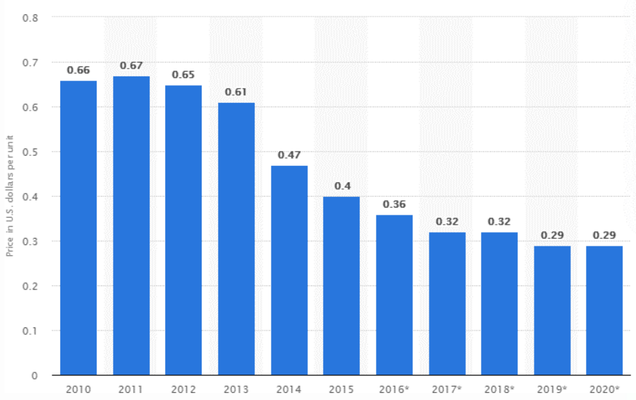


Sensors

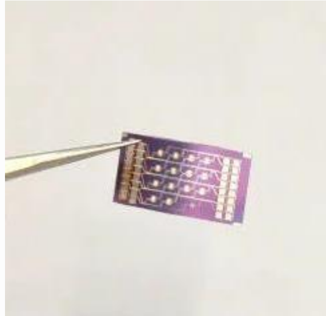


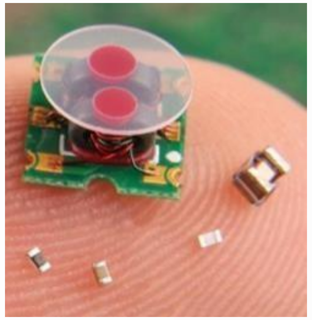
Sensors

There are three primary factors driving the deployment of sensor technology:



- Cheaper sensor
- Smarter sensor
- Smaller sensor



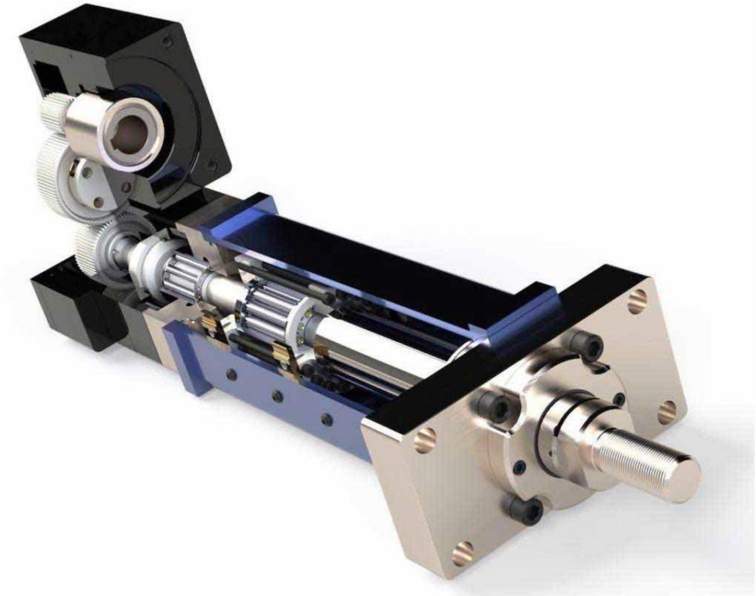


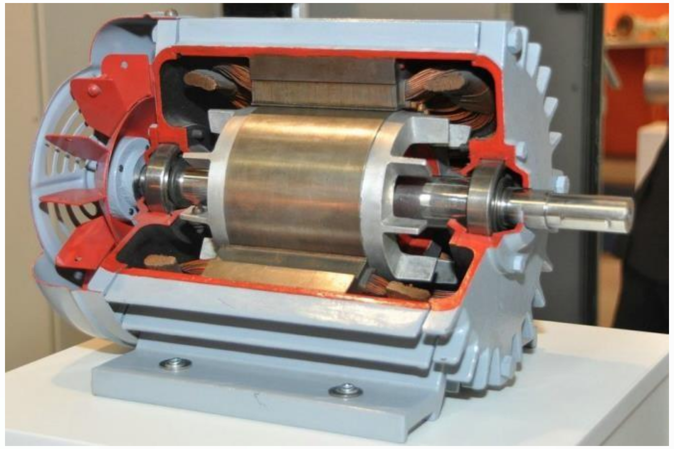
Statista Research Department, Dec 2020

Actuators

Actuators

- Actuator is a device that converts an electrical signal into action, often by converting the signal to nonelectrical energy, such as motion. Example: electric motor
- Actuator types: Hydraulic, Electrical, Thermal, Pneumatic.





Use Cases

Use Case: Smart Diapers

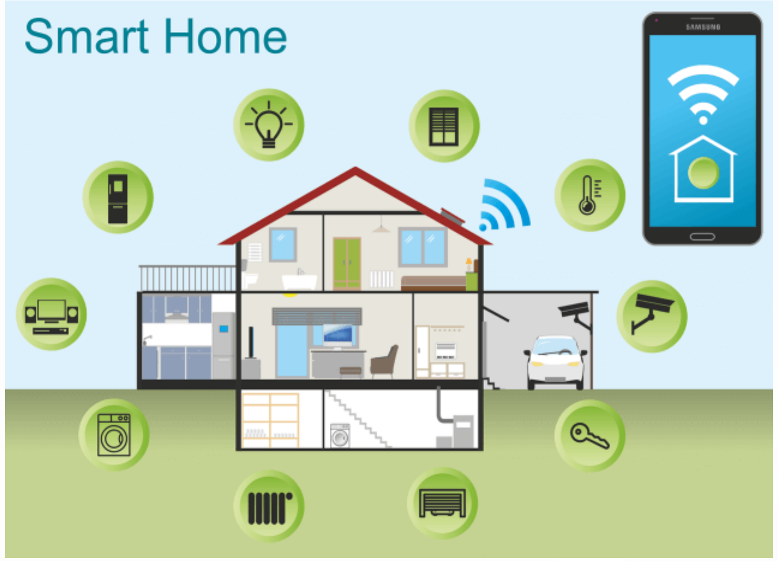
- Diapers have patches at the front with several coloured squares that change colour as they react to different compounds, such as water content, proteins or bacteria.
- Sensors read the data and send notification

Use Case: Smart Clothes

- Hexoskin offers new smart clothing that uses body sensors

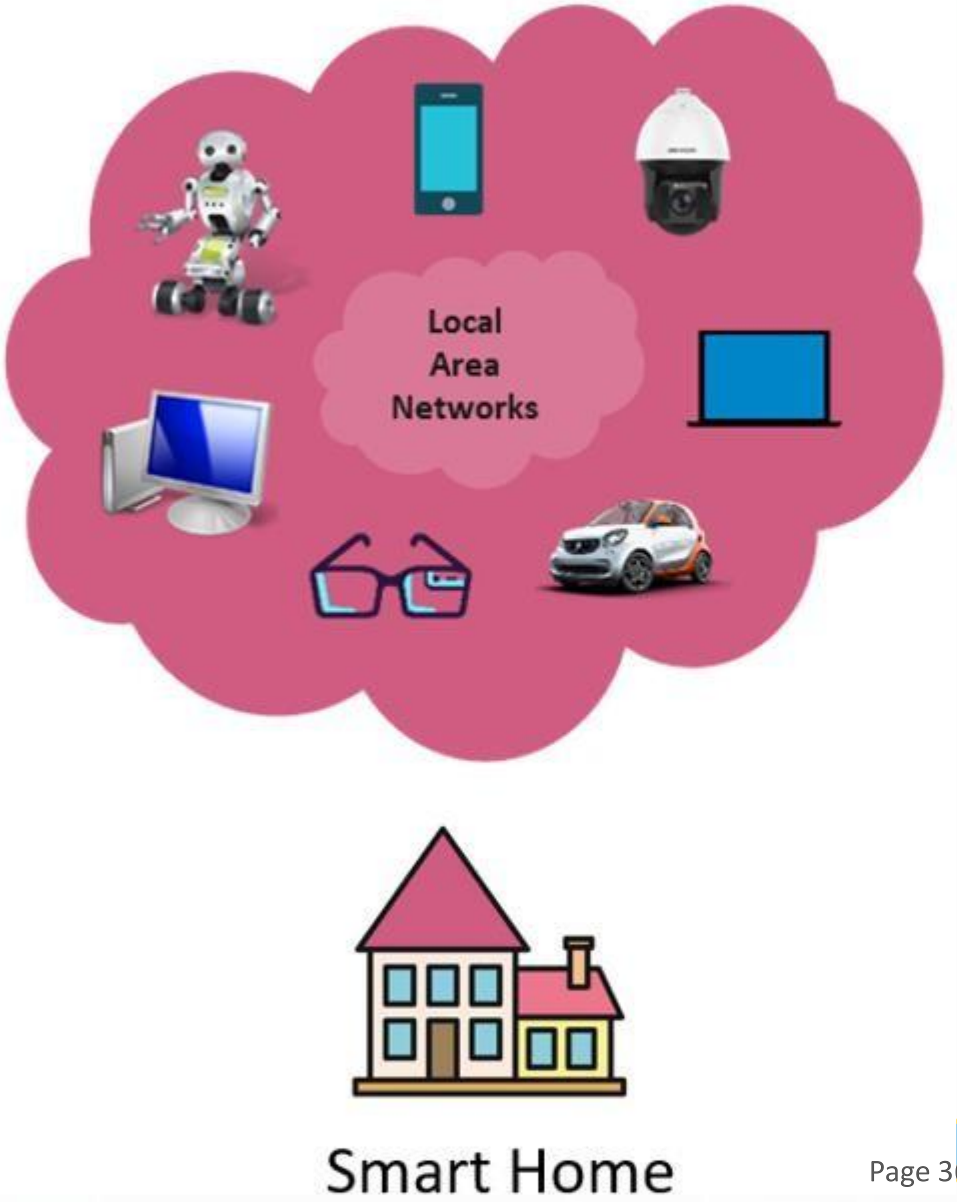
Use Case: Smart Home

- Several environmental, video, audio, and bio sensors are deployed to observe the home environment and physiological health
- Emotion and sentiment detection, activity recognition and situation management are applied to provide healthcare and emergency related services and to manage resources at the home.

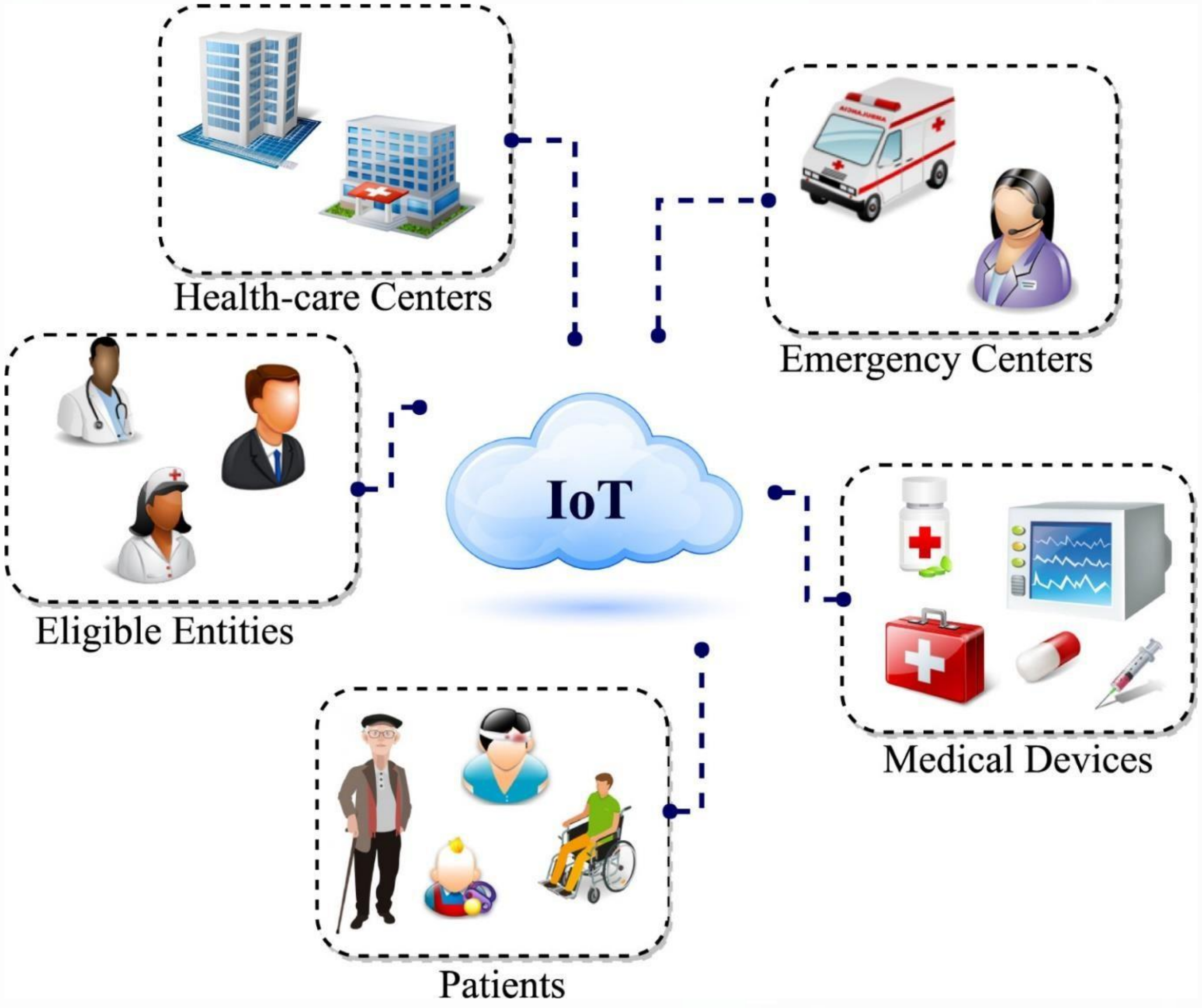


Use Case: Smart Home

- Smart Health
- Smart Ageing
- Automated Video Surveillance
- Energy Management
- Remote Object Manipulation



Use Case: Next Generation Hospital



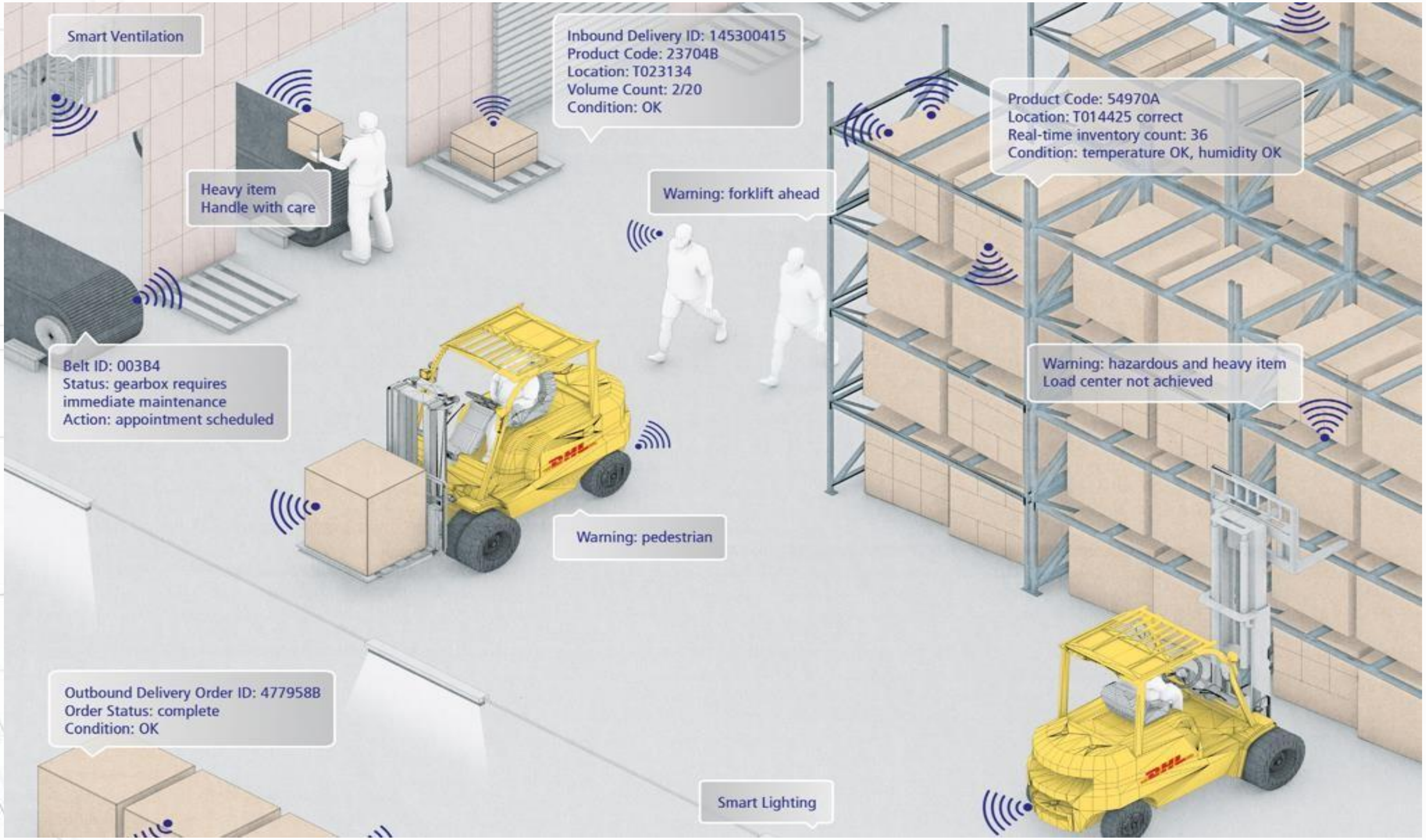
Use Case: Smart Factory

- Interconnected and intelligent manufacturing is competing in the world of Industry 4.0
- Shares data through connected devices and production systems to proactively address issues, improve manufacturing processes.

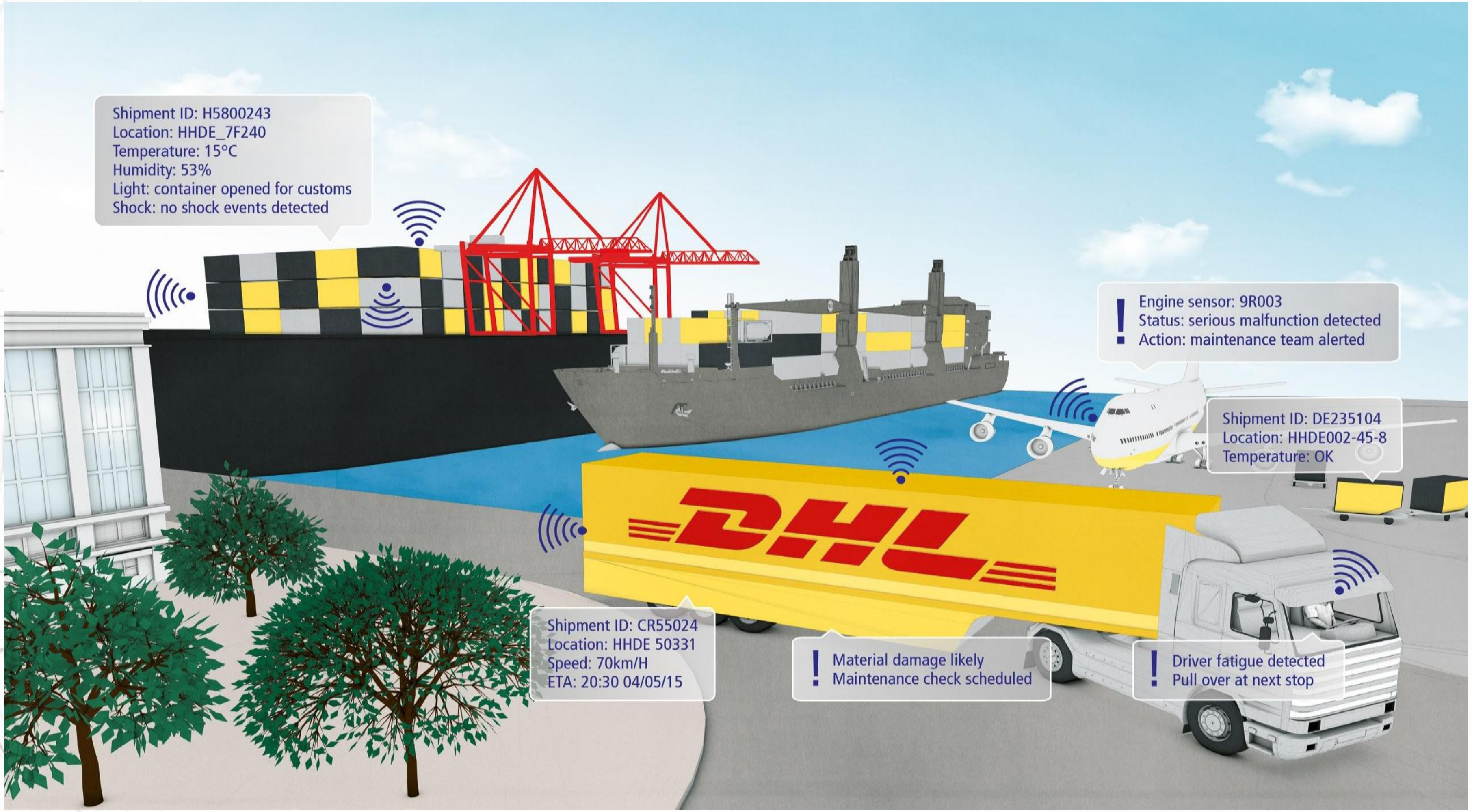
Use Case: Smart Factory

- Monitoring and Control
- Remote Object Manipulation
- Product Packaging
- Product Delivery
- Vehicle Management System

Use Case: Smart Warehouse



Use Case: Freight Transportation



IoT in Logistics: A collaborative report by DHL and Cisco on implications and use cases for the logistics industry

Use Case: Smart Building

- Automated processes to automatically control the building's operations including heating, ventilation, air conditioning, lighting, security and energy

Use Case: Smart Agriculture

- Greenhouse automation;
- Detailed insights into areas;
- Visualisations of real-time data;
- Smart irrigation;
- Disease detection;

Use Case: IoT Based Project

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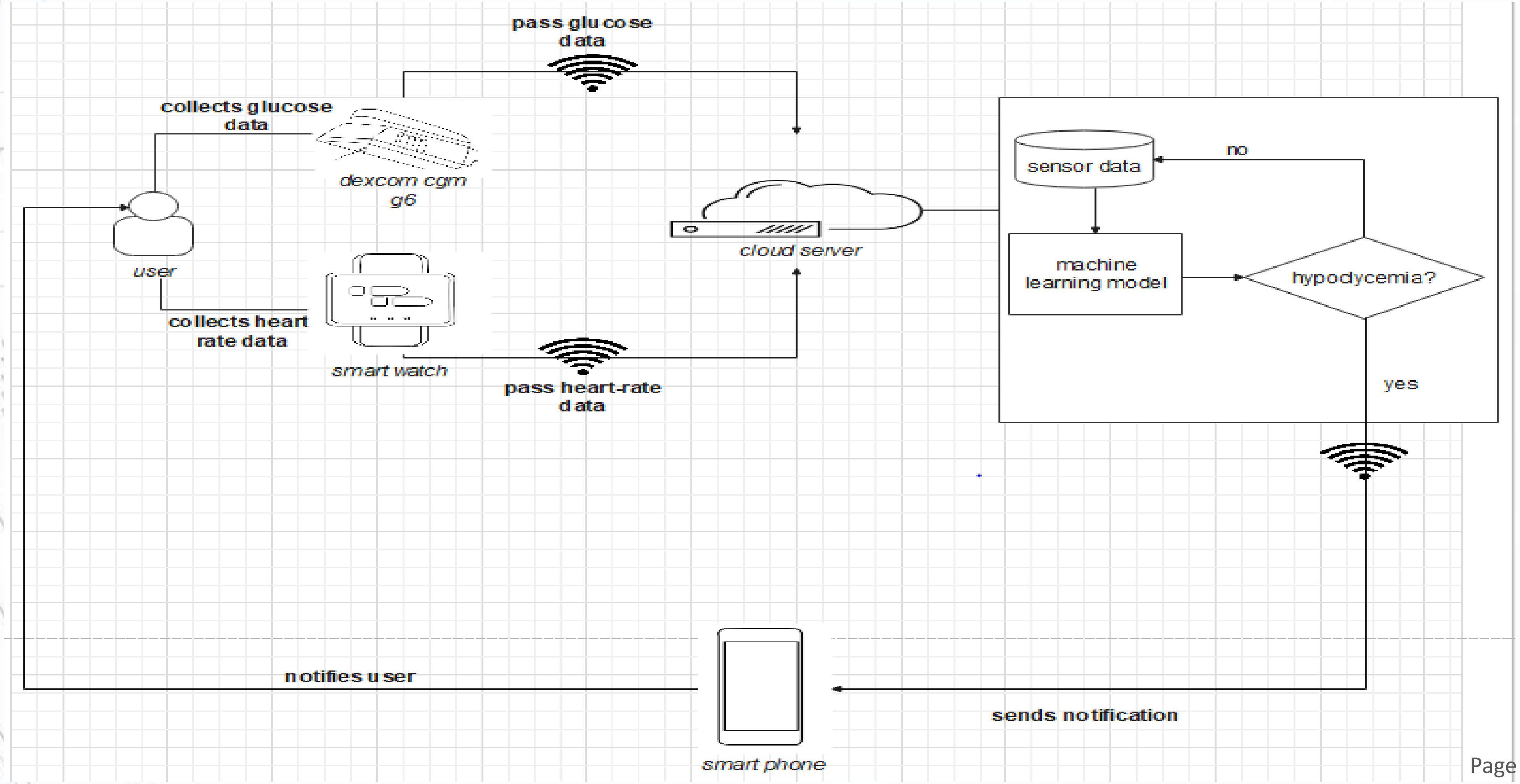
IoT based Hypoglycemia Detection

- Hypoglycemia: Insufficient amount of glucose or sugar in the blood. Blood sugar level goes below 70 mg/dl.

- Data: heart rate, diastolic blood pressure, systolic blood pressure, body temperature, sweating.

IoT based Hypoglycemia Detection

Use Case Diagram



IoT based Hypoglycemia Detection

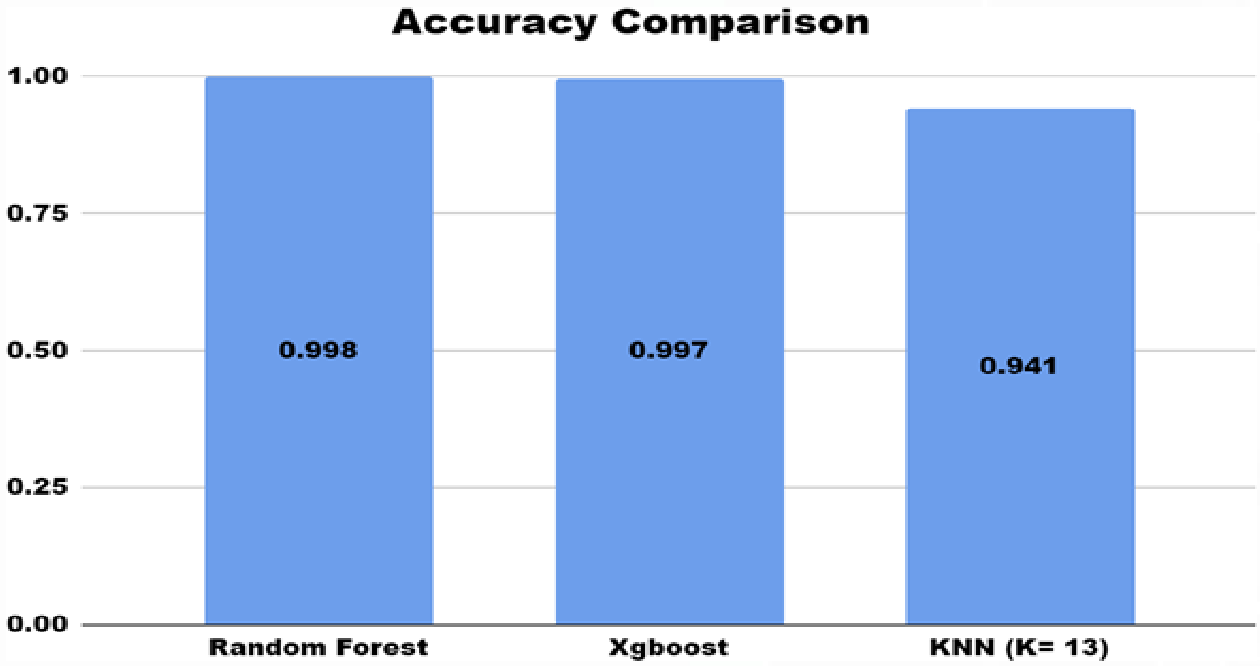
Dataset

	Glucose	Systolic Blood Pressure	Heart Rate	Sweating (Y/N)	Shivering (Y/N)	Hypoglycemia
count	16969.000000	16969.000000	16969.000000	16969.000000	16969.000000	16969.000000
mean	95.741823	118.187165	662.851965	0.121692	0.145501	0.494313
std	42.987604	7.700363	68.677048	0.326940	0.352615	0.499982
min	50.000000	95.000000	461.000000	0.000000	0.000000	0.000000
25%	68.000000	113.000000	631.000000	0.000000	0.000000	0.000000
50%	83.000000	119.000000	674.000000	0.000000	0.000000	0.000000
75%	108.000000	124.000000	714.000000	0.000000	0.000000	1.000000
max	250.000000	145.000000	769.000000	1.000000	1.000000	1.000000

IoT based Hypoglycemia Detection

Three machine learning algorithms were used:

- KNN
- XGBoost
- Random Forest



Outcomes

- Ability to recognize the principals of IoT;
- Ability to determine the challenges of the IoT;
- Ability to determine the advantages and disadvantages of IoT based solutions in our daily life

Ability to think of IoT projects ☺

Thank You

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